

Exercises and references

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Exercises

Exercise 1. — Check that a map

$$\beta: X \times X \rightarrow X \times X, (x, y) \mapsto (x \triangleright y, x)$$

induces actions of the braid groups B_n on the powers X^n of X , i.e.,

$$(\beta \times \text{id})(\text{id} \times \beta)(\beta \times \text{id}) = (\text{id} \times \beta)(\beta \times \text{id})(\text{id} \times \beta)$$

if and only if

$$x \triangleright (y \triangleright z) = (x \triangleright y) \triangleright (x \triangleright z).$$

Exercise 2. — Let $(R, \triangleright)$ be a rack. Show that

$$\sigma(x) = x \triangleright x$$

defines an automorphism of $(R, \triangleright)$.

Exercise 3. — Show that there are, up to isomorphism, six racks with three elements.

Exercise 4. — Let (X, φ) be a permutation rack. Show that $\text{Gr}(X, \varphi)$ is free abelian with basis the set X/φ of orbits.

Exercise 5. — Let $X = R$ be a rack, and $G = \text{Gr}(R)$ its enveloping group. Show that the map $R \rightarrow \text{Gr}(R)$ that sends an element to the corresponding left-multiplication produces a crossed G -set.

Exercise 7. — Verify that

$$(g, s) \triangleright (h, t) = (gs g^{-1}h, t)$$

defines a rack structure on $F(S) \times S$. What is σ for this rack?

Exercise 8. — Given an abelian group M with $s, t: M \rightarrow M$, show that

$$x \triangleright y = sx + ty$$

defines a rack structure on M if $s^2 = s(1 - t)$. Is the converse also true? We want the addition $M \times M \rightarrow M$ to be a rack morphism.

Exercise 9. — Let $\mathbb{Z}$ be the rack with $x \triangleright y = 2x - y$. Is the rack $\mathbb{Z} \times \mathbb{Z}$ free?

Exercise 10. — Check the statements about the complex $\text{HOM}(C, D)$:

$$(\partial f)c = \partial(fc) - (-1)^{|f|}f(\partial c)$$

is a differential. The kernel in degree 0 consists of the homomorphisms, and the image in degree 0 consists of the nullhomotopic homomorphisms $C \rightarrow D$. The zeroth homology is given by the homotopy classes of homomorphisms.

Exercise 11. — The group ring of the infinite cyclic group $\mathbb{Z}$ is isomorphic to the Laurent polynomial ring $\mathbb{Z}[t^{\pm 1}]$. Find a small free resolution of the trivial module $\mathbb{Z}$ (involving only two free modules F_0 and F_1 .) Use that to compute the homology of the infinite cyclic group $\mathbb{Z}$ and give formulas for the (co)homology of all $\mathbb{Z}[t^{\pm 1}]$ -modules M .

Exercise 12. — Given an abelian group A , consider the simplicial abelian group with $A_n = A$ and $\partial_j = \text{id}$ for all j . Find the homology of the associated chain complex.

Exercise 13. — Given a group G , the orbit set of the orbit G/H is a singleton $\star$. What is the homotopy orbit space of G -action on G/H ?

Exercise 14. — Can you describe the homology of the mapping torus of $\varphi: X \rightarrow X$ in terms of the homology $H_\bullet(X)$ and the induced map $H_\bullet(\varphi)$ on it? Test your result with $\varphi = \text{id}$ and other cases of mapping tori that you can work out by other means.

Exercise 15. — Check $\partial^2 = 0$ for the differential on the rack complex. One way to do that is to write $\partial_A = \sum_j (-1)^j A_j$ and similarly $\partial_B = \sum_j (-1)^j B_j$, then you can easily check that the relations

$$\begin{aligned}\partial_A^2 &= 0 \\ \partial_B^2 &= 0 \\ \partial_A \partial_B + \partial_B \partial_A &= 0\end{aligned}$$

hold, and these imply $\partial^2 = (\partial_A - \partial_B)^2 = 0$.

Exercise 16. — Check that a map $\varphi: R \times R \rightarrow A$ defines a rack structure on $R \times A$ via

$$(x, i) \triangleright (y, j) = (x \triangleright y, j + \varphi(x, y))$$

if and only if

$$\varphi(x \triangleright y, x \triangleright z) + \varphi(x, z) = \varphi(x, y \triangleright z) + \varphi(y, z).$$

Exercise 17. — Check the claim about

$$H(x_1, \dots, x_n) = (r, x_1, \dots, x_n) :$$

it is a chain homotopy between $r \triangleright$ and id :

$$(\partial H + H \partial)(x_1, \dots, x_n) = (r \triangleright x_1, \dots, r \triangleright x_n) - (x_1, \dots, x_n)$$

Exercise 18. — In the lecture, we computed the rack homology of the trivial rack. The trivial rack is a quandle. What is its quandle homology?

Exercise 19. — Check, for instance, with a computer and RIG, that the quandle of order 6 given by the 4-cycles in S_4 has homology classes of order 4, so they are not 6-torsion.

Exercise 20. — Describe a bicomplex that has a non-trivial ∂_2 in one of its spectral sequences.

Exercise 21. Can you use the methods of the lectures to show that Quillen (co)homology for G -sets can be identified with the (co)homology of the homotopy orbit space? Or that Quillen (co)homology for groups is ordinary group (co)homology?

Disclaimer

The following is a list of references for the mini-course. It is what I have used in preparation, and it does not in any way constitute a bibliography of the subject. For reasons of time and space, I have been unable to mention many important contributors and contributions, and I am sorry for all those whom I have not been able to represent adequately.

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